Lab Experiment-23

Aim

To write a Python program to evaluate text coherence by measuring the similarity between consecutive sentences using TF-IDF and cosine similarity.

Algorithm

1. Import the required libraries.
2. Get a paragraph from the user.
3. Split the paragraph into sentences.
4. Convert sentences into TF-IDF vectors.
5. Calculate cosine similarity between consecutive sentences.
6. Calculate the average similarity score.
7. Display the coherence score.

Code

```
from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.metrics.pairwise import cosine_similarity

import re

text = input("Enter a paragraph: ")

sentences = re.split(r'[.!?]+', text)

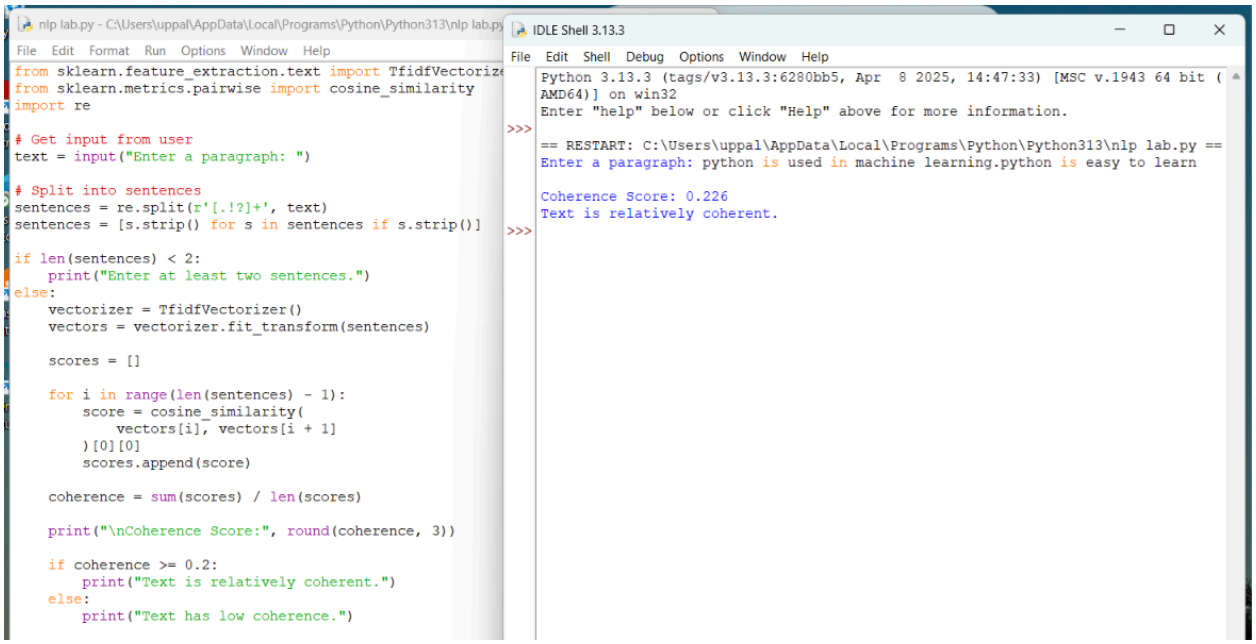
sentences = [s.strip() for s in sentences if s.strip()]

if len(sentences) < 2:

    print("Enter at least two sentences.")
```

```
else:
    vectorizer = TfidfVectorizer()
    vectors = vectorizer.fit_transform(sentences)
    scores = []
    for i in range(len(sentences) - 1):
        score = cosine_similarity(
            vectors[i], vectors[i + 1]
        )[0][0]
        scores.append(score)
    coherence = sum(scores) / len(scores)
    print("\nCoherence Score:", round(coherence, 3))
    if coherence >= 0.2:
        print("Text is relatively coherent.")
    else:
        print("Text has low coherence.")
```

Output:



Result:

Thus, the program successfully evaluates text coherence using TF-IDF and cosine similarity.